In [2]:
```
!pip install numpy
!pip install pandas
!pip install matplotlib.pyplot
!pip install seaborn
!pip install sqlite3
```

```
Collecting numpy
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Installing collected packages: numpy
Successfully installed numpy-2.5.3
Collecting pandas
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s\Python\Python313\Lib\site-packages (from pandas) (2.5.3)
Requirement already satisfied: python-dateutil>=2.8.2 in C:\Users\Arena\AppData\Loca
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Requirement already satisfied: tzdata in C:\Users\Arena\AppData\Local\Programs\Pytho
n\Python313\Lib\site-packages (from pandas) (2026.5)
Requirement already satisfied: six>=1.5 in C:\Users\Arena\AppData\Local\Programs\Pyt
hon\Python313\Lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
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Installing collected packages: pandas
Successfully installed pandas-3.0.6
```

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ERROR: Could not find a version that satisfies the requirement matplotlib.pyplot (fr
om versions: none)
ERROR: No matching distribution found for matplotlib.pyplot
```

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Collecting seaborn
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in C:\Users\Arena\AppData\Local
\Programs\Python\Python313\Lib\site-packages (from seaborn) (2.5.3)
Requirement already satisfied: pandas>=1.2 in C:\Users\Arena\AppData\Local\Programs
\Python\Python313\Lib\site-packages (from seaborn) (3.0.6)
Collecting matplotlib!=3.6.1,>=3.4 (from seaborn)
  Downloading matplotlib-3.11.2-cp313-cp313-win_amd64.whl.metadata (80 kB)
Collecting contourpy>=1.0.1 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading contourpy-1.4.0-cp313-cp313-win_amd64.whl.metadata (3.8 kB)
Collecting cycler>=0.10 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading fonttools-4.66.1-cp313-cp313-win_amd64.whl.metadata (132 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading kiwisolver-1.5.1-cp313-cp313-win_amd64.whl.metadata (5.2 kB)
Requirement already satisfied: packaging>=20.0 in C:\Users\Arena\AppData\Local\Progr
ams\Python\Python313\Lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.
3)
Collecting pillow>=9 (from matplotlib!=3.6.1,>=3.4->seaborn)
  Downloading pillow-12.3.0-cp313-cp313-win_amd64.whl.metadata (9.3 kB)
Collecting pyparsing>=3 (from matplotlib!=3.6.1,>=3.4->seaborn)
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Requirement already satisfied: python-dateutil>=2.7 in C:\Users\Arena\AppData\Local
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Requirement already satisfied: six>=1.5 in C:\Users\Arena\AppData\Local\Programs\Pyt
hon\Python313\Lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4-
>seaborn) (1.17.0)
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Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.66.1-cp313-cp313-win_amd64.whl (2.5 MB)
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Downloading pillow-12.3.0-cp313-cp313-win_amd64.whl (7.2 MB)
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tourpy, matplotlib, seaborn

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```
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   ---------------------------------------- 8/8 [seaborn]

Successfully installed contourpy-1.4.0 cycler-0.12.1 fonttools-4.66.1 kiwisolver-1.
5.1 matplotlib-3.11.2 pillow-12.3.0 pyparsing-3.3.3 seaborn-0.13.2
ERROR: Could not find a version that satisfies the requirement sqlite3 (from version
s: none)
ERROR: No matching distribution found for sqlite3
```

In [3]:
```python
#import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sqlite3
```

In [ ]:
```python
#1. import database / data
```

In [4]:
```python
conn = sqlite3.connect('customer_churn.db')

sql_query = """
            select name
            from sqlite_master
            where type = 'table'
"""

tables = pd.read_sql(sql_query,conn)

# create dataframe for each table

for table_name in tables['name']:
    df = pd.read_sql(f"select * from {table_name}", conn)
    globals()[f"df_{table_name}"] = df
    print(f"Created dataframe: df_{table_name}")

conn.close()
```

```
Created dataframe: df_db_customer
Created dataframe: df_db_subscription
Created dataframe: df_db_support
```

In [5]:
```python
# print table names and column names
conn = sqlite3.connect('customer_churn.db')
```

```
for table_name in tables['name']:
    print(f"\nTable Name: {table_name}")
    # Get Column information
    columns_query = f"PRAGMA table_info({table_name});"
    columns = pd.read_sql(columns_query , conn)
    print("columns:")
    print(columns['name'].tolist())
# close connection
conn.close()
```

```
Table Name: db_customer
columns:
['customerid', 'name', 'country', 'state', 'gender', 'dob', 'interests', 'pincode']

Table Name: db_subscription
columns:
['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'plan
_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_charge
s', 'cltv', 'churn_score']

Table Name: db_support
columns:
['customerid', 'complaint_date', 'escalations', 'csat_score', 'col_1', 'comment']
```

In [ ]:
```
# data cleaning
```

In [6]:
```
df_db_customer.head()
```

Out[6]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>interests</th> <th>pincode</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

| | customerid | name | country | state | gender | dob | interests | pincode |
|---|---|---|---|---|---|---|---|---|
| 0 | 0002-ORFBO | keshav | India | Maharashtra | Male | 1982-04-12 00:00:00 | travel | None |
| 1 | 0003-MKNFE | raghav | India | Karnataka | Male | 1995-11-23 00:00:00 | NaN | None |
| 2 | 0004-TLHLJ | lalita | India | Delhi | Female | 1978-02-15 00:00:00 | movie | None |
| 3 | 0011-IGKFF | mohan | India | Nagaland | Male | 2001-08-30 00:00:00 | NaN | None |
| 4 | 0013-EXCHZ | mira | India | Delhi | Female | 1990-05-05 00:00:00 | drama | None |

In [8]:
```
df_db_customer.tail()
```

Out[8]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>interests</th> <th>pincode</th> </thead> <tbody> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0020-JDNXP | rikim | India | Meghalaya | Female | 1994-08-19 00:00:00 | NaN | None |
| 0021-IKXGC | vishakha | India | Rajasthan | Female | 2000-09-02 00:00:00 | NaN | None |
| 0022-TCJCI | raghvendra | India | Telangana | Male | 1983-12-30 00:00:00 | NaN | None |
| 0023-HGHWL | rishabh | India | Uttar Pradesh | Men | 1991-05-14 00:00:00 | NaN | None |
| 0023-UYUPN | sudevi | India | Maharashtra | Women | 1977-10-06 00:00:00 | NaN | None |

In [7]:
```
df_db_customer.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   customerid  21 non-null     str
 1   name        21 non-null     str
 2   country     18 non-null     str
 3   state       21 non-null     str
 4   gender      21 non-null     str
 5   dob         21 non-null     str
 6   interests   4 non-null      str
 7   pincode     0 non-null      object
dtypes: object(1), str(7)
memory usage: 1.4+ KB
```

In [ ]:
```
# a. rename column - name to customer_name
# b. drop columns - interest and pincode
# c. change datatype - dob
# d. data standardization - gender
# e. fix missing values - country
```

In [14]:
```
# a. rename col - name

df_db_customer.rename(columns = {'name' : 'customer_name'}, inplace = True)
df_db_customer
```

Out[14]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>interests</th> <th>pincode</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0002-ORFBO | keshav | India | Maharashtra | Male | 1982-04-12 00:00:00 | travel | None |
| 0003-MKNFE | raghav | India | Karnataka | Male | 1995-11-23 00:00:00 | NaN | None |
| 0004-TLHLJ | lalita | India | Delhi | Female | 1978-02-15 00:00:00 | movie | None |
| 0011-IGKFF | mohan | India | Nagaland | Male | 2001-08-30 00:00:00 | NaN | None |
| 0013-EXCHZ | mira | India | Delhi | Female | 1990-05-05 00:00:00 | drama | None |
| 0013-MHZWF | durga | NaN | Delhi | Women | 1988-12-10 00:00:00 | NaN | None |
| 0013-SMEOE | mina | India | Meghalaya | Female | 1976-09-21 00:00:00 | NaN | None |
| 0014-BMAQU | madan | India | Rajasthan | Male | 1999-03-14 00:00:00 | NaN | None |
| 0015-UOCOJ | maya | NaN | Kathmandu | Women | 1985-07-07 00:00:00 | NaN | None |
| 0016-QLJIS | arjun | Nepal | Kathmandu | Male | 1993-10-29 00:00:00 | NaN | None |
| 0017-DINOC | shiva | India | Maharashtra | Men | 1997-01-22 00:00:00 | NaN | None |
| 0017-IUDMW | rangadevi | India | Karnataka | Female | 1981-06-18 00:00:00 | NaN | None |
| 0018-NYROU | chitra | NaN | Telangana | Female | 2004-12-01 00:00:00 | NaN | None |
| 0019-EFAEP | raju | India | Meghalaya | Female | 1992-04-25 00:00:00 | NaN | None |
| 0019-GFNTW | Madhav | India | Uttar Pradesh | Men | 1979-11-11 00:00:00 | NaN | None |
| 0020-INWCK | parvati | India | Delhi | Female | 1986-02-28 00:00:00 | job | None |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0020-JDNXP | rikim | India | Meghalaya | Female | 1994-08-19 00:00:00 | NaN | None |
| 0021-IKXGC | vishakha | India | Rajasthan | Female | 2000-09-02 00:00:00 | NaN | None |
| 0022-TCJCI | raghvendra | India | Telangana | Male | 1983-12-30 00:00:00 | NaN | None |
| 0023-HGHWL | rishabh | India | Uttar Pradesh | Men | 1991-05-14 00:00:00 | NaN | None |
| 0023-UYUPN | sudevi | India | Maharashtra | Women | 1977-10-06 00:00:00 | NaN | None |

In [25]:
```
# b. drop columns - interest and pincode

df_db_customer.drop(df_db_customer.columns[-2:], axis = 1)
#df_db_customer.drop(df_db_customer.columns[6:], axis = 1)

#df_db_customer.drop(columns=['interests','pincode'],inplace=True)
```

Out[25]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody></tbody>

| | | | |
|---|---|---|---|
| 0002-ORFBO | keshav | India | Maharashtra |
| 0003-MKNFE | raghav | India | Karnataka |
| 0004-TLHLJ | lalita | India | Delhi |
| 0011-IGKFF | mohan | India | Nagaland |
| 0013-EXCHZ | mira | India | Delhi |
| 0013-MHZWF | durga | NaN | Delhi |
| 0013-SMEOE | mina | India | Meghalaya |
| 0014-BMAQU | madan | India | Rajasthan |
| 0015-UOCOJ | maya | NaN | Kathmandu |
| 0016-QLJIS | arjun | Nepal | Kathmandu |
| 0017-DINOC | shiva | India | Maharashtra |
| 0017-IUDMW | rangadevi | India | Karnataka |
| 0018-NYROU | chitra | NaN | Telangana |
| 0019-EFAEP | raju | India | Meghalaya |
| 0019-GFNTW | Madhav | India | Uttar Pradesh |
| 0020-INWCK | parvati | India | Delhi |
| 0020-JDNXP | rikim | India | Meghalaya |
| 0021-IKXGC | vishakha | India | Rajasthan |
| 0022-TCJCI | raghvendra | India | Telangana |
| 0023-HGHWL | rishabh | India | Uttar Pradesh |
| 0023-UYUPN | sudevi | India | Maharashtra |

In [29]:
```
df_db_customer.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   customerid     21 non-null     str
 1   customer_name  21 non-null     str
 2   country        18 non-null     str
 3   state          21 non-null     str
 4   gender         21 non-null     str
 5   dob            21 non-null     datetime64[us]
dtypes: datetime64[us](1), str(5)
memory usage: 1.1 KB
```

In [28]:
```
# c. change datatype - dob

df_db_customer['dob'] = pd.to_datetime(df_db_customer['dob'])
```

In [31]:
```
# d. data standardization - gender

#df_db_customer['gender'].unique()
df_db_customer['gender'] = df_db_customer['gender'].replace({'Men' : 'Male' , 'Wome
```

In [32]:
```
df_db_customer['gender'].unique()
```

Out[32]:
```
<StringArray>
['Male', 'Female']
Length: 2, dtype: str
```

In [34]:
```
# e. fix missing values - country

df_db_customer[df_db_customer['country'].isna()]
```

Out[34]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> <th>5</th> <th>8</th> <th>12</th> </tbody> <tbody></tbody>

| | | | | | | |
|---|---|---|---|---|---|---|
| 0013-MHZWF | durga | NaN | Delhi | Female | 1988-12-10 |
| 0015-UOCOJ | maya | NaN | Kathmandu | Female | 1985-07-07 |
| 0018-NYROU | chitra | NaN | Telangana | Female | 2004-12-01 |

In [40]:
```
# country and state - unique value pair

state_country_mapping =  df_db_customer.dropna(subset=['country']).set_index('state

df_db_customer['country'] = df_db_customer['country'].fillna(df_db_customer['state'
```

In [41]:
```
df_db_customer[df_db_customer['country'].isna()]
```

Out[41]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> </thead> <tbody> </tbody> <tbody></tbody>

In [ ]:

In [42]:
```
df_db_subscription.head()
```

Out[42]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>subscription_start_date</th> <th>subscription_type</th> <th>renewal_date</th> <th>plan_type</th> <th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th> <th>monthly_charges</th> <th>cltv</th> <th>churn_score</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual | NaN | NaN | 13.99 | 627 | 12 |
| 0003-MKNFE | 2020-08-01 | Paid | 2024-08-01 | Premium | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 |
| 0004-TLHLJ | 2022-11-20 | Organic | 2025-11-20 | Basic | Monthly | NaN | NaN | 6.99 | 210 | 34 |
| 0011-IGKFF | 2019-05-10 | Paid | 2025-05-10 | Premium | Annual | NaN | NaN | 22.99 | 1725 | 8 |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive | 13.99 | 195 | 88 |

In [43]:
```
df_db_subscription.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customerid              21 non-null     str
 1   subscription_start_date  21 non-null     str
 2   subscription_type       21 non-null     str
 3   renewal_date            21 non-null     str
 4   plan_type               21 non-null     str
 5   contract_type           21 non-null     str
 6   cancellation_date       6 non-null      str
 7   cancellation_reason     6 non-null      str
 8   monthly_charges         21 non-null     float64
 9   cltv                    21 non-null     int64
 10  churn_score             21 non-null     int64
dtypes: float64(1), int64(2), str(8)
memory usage: 1.9 KB
```

In [47]:
```
date_col = ['subscription_start_date', 'renewal_date' , 'cancellation_date']

df_db_subscription[date_col] = df_db_subscription[date_col].apply(pd.to_datetime)
df_db_subscription.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customerid              21 non-null     str
 1   subscription_start_date  21 non-null     datetime64[us]
 2   subscription_type       21 non-null     str
 3   renewal_date            21 non-null     datetime64[us]
 4   plan_type               21 non-null     str
 5   contract_type           21 non-null     str
 6   cancellation_date       6 non-null      datetime64[us]
 7   cancellation_reason     6 non-null      str
 8   monthly_charges         21 non-null     float64
 9   cltv                    21 non-null     int64
 10  churn_score             21 non-null     int64
dtypes: datetime64[us](3), float64(1), int64(2), str(5)
memory usage: 1.9 KB
```

In [ ]:
```
```

In [48]:
```
df_db_support.head()
```

Out[48]:
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> <th>col_1</th> <th>comment</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

| | | | | | | |
|---|---|---|---|---|---|---|
| 0003-MKNFE | 2024-08-28 00:00:00 | N | 60 | None | service issue |
| 0003-MKNFE | 2024-08-28 00:00:00 | Y | 10 | None | demaned refund |
| 0013-EXCHZ | 2024-01-20 00:00:00 | Y | 20 | None | NaN |
| 0013-MHZWF | 2025-03-18 00:00:00 | N | 90 | None | guidance to renew |
| 0013-SMEOE | 2024-11-01 00:00:00 | N | 30 | None | NaN |

In [54]:
```
df_db_support.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   customerid      9 non-null      str
 1   complaint_date  9 non-null      datetime64[us]
 2   escalations     9 non-null      str
 3   csat_score      9 non-null      int64
dtypes: datetime64[us](1), int64(1), str(2)
memory usage: 420.0 bytes
```

In [51]:
```
df_db_support.drop(columns=['col_1','comment'],inplace=True)
```

In [53]:
```
df_db_support['complaint_date'] = pd.to_datetime(df_db_support['complaint_date'])

df_db_support.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   customerid      9 non-null      str
 1   complaint_date  9 non-null      datetime64[us]
 2   escalations     9 non-null      str
 3   csat_score      9 non-null      int64
dtypes: datetime64[us](1), int64(1), str(2)
memory usage: 420.0 bytes
```

In [ ]:
```

```

In [55]:
```
#3. Feature Engineering and Data Analysis
```

In [56]:
```
#create a new col using existing col - churn flag
df_db_subscription['churn_flag'] = np.where(df_db_subscription['cancellation_date']
```

In [57]:
```
df_db_subscription.head()
```

Out[57]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>subscription_start_date</th> <th>subscription_type</th> <th>renewal_date</th> <th>plan_type</th> <th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th> <th>monthly_charges</th> <th>cltv</th> <th>churn_score</th> <th>churn_flag</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual | NaT | NaN | 13.99 | 627 | 12 | 0 |
| 0003-MKNFE | 2020-08-01 | Paid | 2024-08-01 | Premium | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | 91 | 1 |
| 0004-TLHLJ | 2022-11-20 | Organic | 2025-11-20 | Basic | Monthly | NaT | NaN | 6.99 | 210 | 34 | 0 |
| 0011-IGKFF | 2019-05-10 | Paid | 2025-05-10 | Premium | Annual | NaT | NaN | 22.99 | 1725 | 8 | 0 |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive | 13.99 | 195 | 88 | 1 |

In [60]:
```
df = (df_db_subscription
            .merge(df_db_customer,on = 'customerid' , how = 'left')
            .merge(df_db_support,on = 'customerid' , how = 'left'))
```

In [61]:
```
df_db_subscription.shape
```

Out[61]: (21, 12)

In [62]:
```
df.shape
```

Out[62]: (23, 20)

In [64]:
```
df_db_subscription['customerid'].unique()
```

Out[64]:
```
<StringArray>
['0002-ORFBO', '0003-MKNFE', '0004-TLHLJ', '0011-IGKFF', '0013-EXCHZ',
 '0013-MHZWF', '0013-SMEOE', '0014-BMAQU', '0015-UOCOJ', '0016-QLJIS',
 '0017-DINOC', '0017-IUDMW', '0018-NYROU', '0019-EFAEP', '0019-GFNTW',
 '0020-INWCK', '0020-JDNXP', '0021-IKXGC', '0022-TCJCI', '0023-HGHWL',
 '0023-UYUPN']
Length: 21, dtype: str
```

In [65]:
```
df_db_customer['customerid'].unique()
```

Out[65]:
```
<StringArray>
['0002-ORFBO', '0003-MKNFE', '0004-TLHLJ', '0011-IGKFF', '0013-EXCHZ',
 '0013-MHZWF', '0013-SMEOE', '0014-BMAQU', '0015-UOCOJ', '0016-QLJIS',
 '0017-DINOC', '0017-IUDMW', '0018-NYROU', '0019-EFAEP', '0019-GFNTW',
 '0020-INWCK', '0020-JDNXP', '0021-IKXGC', '0022-TCJCI', '0023-HGHWL',
 '0023-UYUPN']
Length: 21, dtype: str
```

In [66]:
```
df_db_support['customerid'].unique()
```

Out[66]:
```
<StringArray>
['0003-MKNFE', '0013-EXCHZ', '0013-MHZWF', '0013-SMEOE', '0017-IUDMW',
 '0019-EFAEP', '0022-TCJCI']
Length: 7, dtype: str
```

In [67]:
```
df_db_support['customerid'].size
```

Out[67]: 9

In [69]:
```
df_db_support
```

Out[69]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> </tbody> <tbody> </tbody>

| | | | | |
|---|---|---|---|---|
| 0003-MKNFE | 2024-08-28 | N | 60 | |
| 0003-MKNFE | 2024-08-28 | Y | 10 | |
| 0013-EXCHZ | 2024-01-20 | Y | 20 | |
| 0013-MHZWF | 2025-03-18 | N | 90 | |
| 0013-SMEOE | 2024-11-01 | N | 30 | |
| 0017-IUDMW | 2024-04-10 | Y | 25 | |
| 0019-EFAEP | 2024-09-27 | Y | 30 | |
| 0022-TCJCI | 2024-09-13 | Y | 10 | |
| 0022-TCJCI | 2024-09-14 | N | 90 | |

In [72]:
```
df_db_support['complaint_count'] = df_db_support.groupby('customerid')['customerid'
```

In [73]:
```
df_db_support
```

Out[73]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> <th>complaint_count</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> </tbody> <tbody></tbody>

| | | | | | |
|---|---|---|---|---|---|
| 0003-MKNFE | 2024-08-28 | N | 60 | 2 | |
| 0003-MKNFE | 2024-08-28 | Y | 10 | 2 | |
| 0013-EXCHZ | 2024-01-20 | Y | 20 | 1 | |
| 0013-MHZWF | 2025-03-18 | N | 90 | 1 | |
| 0013-SMEOE | 2024-11-01 | N | 30 | 1 | |
| 0017-IUDMW | 2024-04-10 | Y | 25 | 1 | |
| 0019-EFAEP | 2024-09-27 | Y | 30 | 1 | |
| 0022-TCJCI | 2024-09-13 | Y | 10 | 2 | |
| 0022-TCJCI | 2024-09-14 | N | 90 | 2 | |

In [75]:
```
df_db_support = df_db_support.sort_values('complaint_date').drop_duplicates('custom
```

In [76]:
```
df_db_support['customerid'].size
```

Out[76]: 7

In [157…
```
df = (df_db_subscription
            .merge(df_db_customer,on = 'customerid' , how = 'left')
            .merge(df_db_support,on = 'customerid' , how = 'left'))
```

In [158…
```
df.shape
```

Out[158… (21, 21)

In [82]:
```
df.to_csv('exported_churn_data.csv', index=False)
```

In [159…
```
df
```

Out[159…

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>customerid</th> <th>subscription_start_date</th> <th>subscription_type</th> <th>renewal_date</th> <th>plan_type</th> <th>contract_type</th> <th>cancellation_date</th> <th>cancellation_reason</th> <th>monthly_charges</th> <th>cltv</th> <th>...</th> <th>churn_flag</th> <th>customer_name</th> <th>country</th> <th>state</th> <th>gender</th> <th>dob</th> <th>complaint_date</th> <th>escalations</th> <th>csat_score</th> <th>complaint_count</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th> <th>12</th> <th>13</th> <th>14</th> <th>15</th> <th>16</th> <th>17</th> <th>18</th> <th>19</th> <th>20</th> </tbody> <tbody> </tbody>

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0002-ORFBO | 2021-03-15 | Refferal | 2025-03-15 | Standard | Annual | NaT | NaN | 13.99 | 627 | ... | 0 |
| 0003-MKNFE | 2020-08-01 | Paid | 2024-08-01 | Premium | Annual | 2024-09-10 | Switched to competitor | 12.99 | 1150 | ... | 1 |
| 0004-TLHLJ | 2022-11-20 | Organic | 2025-11-20 | Basic | Monthly | NaT | NaN | 6.99 | 210 | ... | 0 |
| 0011-IGKFF | 2019-05-10 | Paid | 2025-05-10 | Premium | Annual | NaT | NaN | 22.99 | 1725 | ... | 0 |
| 0013-EXCHZ | 2023-01-05 | Refferal | 2024-01-05 | Standard | Monthly | 2024-02-28 | Too expensive | 13.99 | 195 | ... | 1 |
| 0013-MHZWF | 2022-06-18 | Paid | 2025-06-18 | Standard | Annual | NaT | NaN | 17.99 | 720 | ... | 0 |
| 0013-SMEOE | 2021-09-30 | Refferal | 2024-09-30 | Basic | Monthly | 2024-11-15 | Not enough content | 8.99 | 230 | ... | 1 |
| 0014-BMAQU | 2020-02-14 | Organic | 2025-02-14 | Premium | Annual | NaT | NaN | 22.99 | 1840 | ... | 0 |
| 0015-UOCOJ | 2023-07-22 | Organic | 2024-07-22 | Standard | Monthly | NaT | NaN | 13.99 | 240 | ... | 0 |
| 0016-QLJIS | 2022-04-03 | Organic | 2025-04-03 | Basic | Annual | NaT | NaN | 6.99 | 335 | ... | 0 |
| 0017-DINOC | 2021-12-01 | Organic | 2025-12-01 | Premium | Annual | NaT | NaN | 22.99 | 1610 | ... | 0 |
| 0017-IUDMW | 2023-03-17 | Refferal | 2024-03-17 | Standard | Monthly | 2024-05-01 | Poor streaming quality | 7.99 | 270 | ... | 1 |
| 0018-NYROU | 2020-10-09 | Organic | 2025-10-09 | Standard | Annual | NaT | NaN | 13.99 | 980 | ... | 0 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0019-EFAEP | 2022-08-25 | Refferal | 2024-08-25 | Basic | Monthly | 2024-10-31 | Switched to competitor | 12.99 | 160 | ... | 1 |
| 0019-GFNTW | 2019-11-11 | Paid | 2025-11-11 | Premium | Annual | NaT | NaN | 92.99 | 2185 | ... | 0 |
| 0020-INWCK | 2023-05-06 | Organic | 2025-05-06 | Standard | Monthly | NaT | NaN | 13.99 | 640 | ... | 0 |
| 0020-JDNXP | 2022-01-19 | Organic | 2024-01-19 | Premium | Annual | NaT | NaN | 20.99 | 550 | ... | 0 |
| 0021-IKXGC | 2021-07-07 | Paid | 2025-07-07 | Standard | Annual | NaT | NaN | 13.99 | 840 | ... | 0 |
| 0022-TCJCI | 2023-09-14 | Refferal | 2024-09-14 | Basic | Monthly | 2024-09-14 | Forgot to cancel trial | 16.99 | 42 | ... | 1 ra |
| 0023-HGHWL | 2020-06-23 | Organic | 2025-06-23 | Premium | Annual | NaT | NaN | 22.99 | 1955 | ... | 0 |
| 0023-UYUPN | 2022-12-31 | Paid | 2025-12-31 | Standard | Monthly | NaT | NaN | 13.99 | 790 | ... | 0 |

21 rows × 21 columns

In [80]:
```
#data analysis
```

In [83]:
```
#1. churn rate
```

In [84]:
```
df.columns
```

Out[84]:
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count'],
      dtype='str')
```

In [87]:
```
churn_rate = df['churn_flag'].mean()*100
print("Churn Rate = ", round(churn_rate,2),"%")
```

```
Churn Rate =  28.57 %
```

In [88]:
```
#2.retenion rate
retention_rate = 100 - churn_rate
print("Retention Rate = ", round(retention_rate,2),"%")
```

```
Retention Rate =  71.43 %
```

In [92]:
```
#3. churn by plan type
churn_by_plan = df.groupby('plan_type')['churn_flag'].mean().mul(100).round(2).rese

print(churn_by_plan)
```

```
  plan_type  churn_rate_pct
0     Basic           60.00
1   Premium           14.29
2  Standard           22.22
```

In [ ]:
```
#4.a. churn by state + sum(revenue) & count of users
#4.b. churn by subscription type + sum(revenue) & count of users
```

In [95]:
```
#5. ARPU - Avg Revenue per User
arpu = df['monthly_charges'].mean()
print('ARPU =', round(arpu,2))
```

```
ARPU = 18.85
```

In [189…
```
#6.Avg Customer Tenure

today = pd.Timestamp.today()

df['tenure_days'] = np.where(
         df['cancellation_date'].notna(),
    (df['cancellation_date'] - df['subscription_start_date']).dt.days ,
    (today - df['subscription_start_date']).dt.days

)

avg_tenure = df['tenure_days'].mean()
print("Avg Tenure (Days) =",round(avg_tenure),0)
```

```
Avg Tenure (Days) = 1557 0
```

In [191…
```
df.columns
```

Out[191…
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'churn_risk', 'tenure_days'],
      dtype='str')
```

In [152…
```
#7. Revenue at risk - revenue lost from churned users
revenue_at_risk = df.loc[df['churn_flag']==1 , 'monthly_charges'].sum()
print("Revenue at  Risk(Rs 'K') =",revenue_at_risk)
```

```
Revenue at  Risk(Rs 'K') = 73.94
```

In [163…
```
#8. Escalation Rate
escalation_rate = (df['escalations']== 'Y').mean()*100
print("Escalation Rate =", round(escalation_rate , 2),"%")
```

```
Escalation Rate = 19.05 %
```

In [172…
```
#9.Avg Complaint Rate per User
avg_complaint = df['complaint_count'].sum()/df['customerid'].nunique()
print("Avg Complaint per User =",round(avg_complaint,2))
```

```
Avg Complaint per User = 0.43
```

In [179…
```
#10. correlation Escalation vs Churn
corr_df = df[['escalations','churn_flag']].dropna()

#df['escalations'] =np.where(df['escalations'] == 'Y',1,0)

correlation = corr_df['escalations'].corr( df['churn_flag'])
print('Correlation between escalation vs churn is =',round(correlation,2))
```

```
Correlation between escalation vs churn is = 0.77
```

In [181…
```
#11. Churn risk - create a column using existing col
conditions =[
             (df['churn_score']<50),
             (df['churn_score']>=50) & (df['churn_score']<70),
             (df['churn_score']>=70)
]

choices = ['low' , 'med' , 'high']
df['churn_risk'] = np.select(conditions , choices , default='unknown')
```

In [182…
```
df[['churn_risk','churn_score']].head()
```

Out[182…

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>churn_risk</th> <th>churn_score</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

| | churn_risk | churn_score |
|---|---|---|
| 0 | low | 12 |
| 1 | high | 91 |
| 2 | low | 34 |
| 3 | low | 8 |
| 4 | high | 88 |

In [ ]:
```
#$. visualization using matplotlib
```

In [194…
```
df_visual = df.copy()
```

In [195…
```
df_visual.shape
```

Out[195…
```
(21, 23)
```

In [192…
```
df.columns
```

Out[192…
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'churn_risk', 'tenure_days'],
      dtype='str')
```

In [196…
```
df_visual.columns
```

Out[196…
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'churn_risk', 'tenure_days'],
      dtype='str')
```

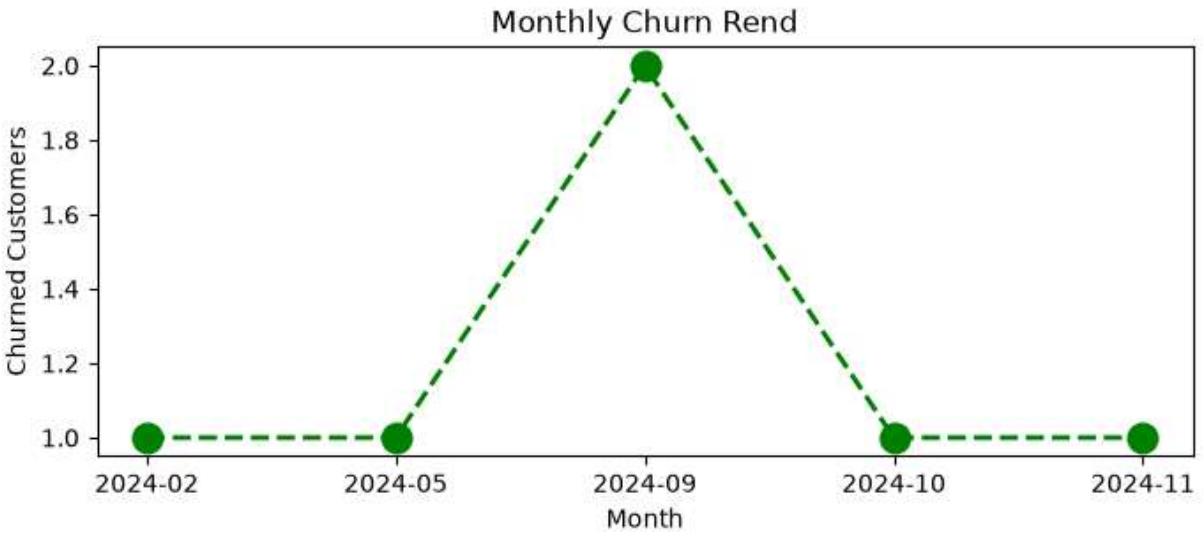
In [202…
```
#4.1 Monthly Churn  (Time Series KPI)

df_visual['cancellation_month'] = df_visual['cancellation_date'].dt.to_period('M')

churn_trend = df_visual[df_visual['churn_flag'] ==1].groupby('cancellation_month').

plt.figure(figsize=(8,3))
plt.plot(churn_trend.index.astype(str),churn_trend.values,color='green', marker='o'

plt.title('Monthly Churn Rend')
plt.xlabel('Month')
plt.ylabel('Churned Customers')
plt.show()
```

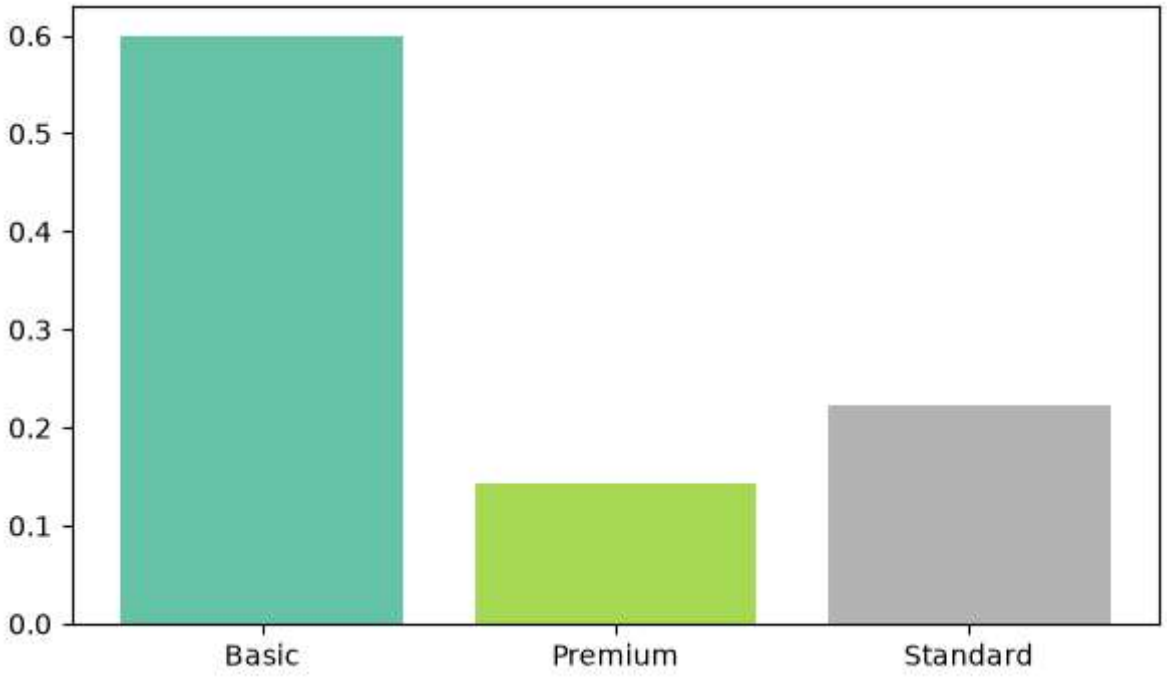



In [209…
```
#4.2 Churn by Plan type
churn_plan = df_visual.groupby('plan_type')['churn_flag'].mean()

#colors = ['yellow','purple','blue']
colors = plt.cm.Set2(np.linspace(0,1,len(churn_plan)))

plt.figure(figsize=(7,4))

plt.bar(churn_plan.index,churn_plan.values,color = colors)
plt.show()
```



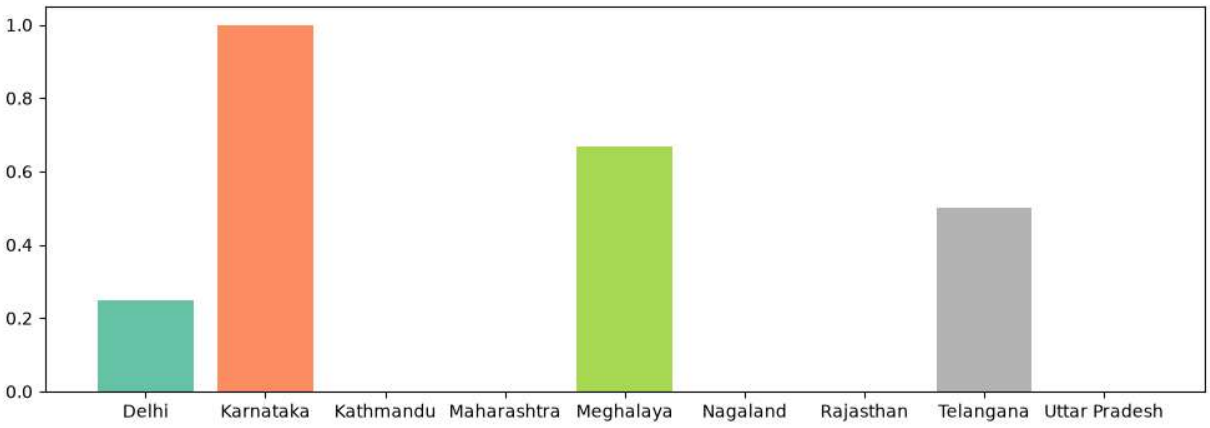

In [210…
```
#4.3 churn by states

churn_plan = df_visual.groupby('state')['churn_flag'].mean()

#colors = ['yellow','purple','blue']
colors = plt.cm.Set2(np.linspace(0,1,len(churn_plan)))

plt.figure(figsize=(12,4))

plt.bar(churn_plan.index,churn_plan.values,color = colors)
plt.show()
```




In [211…
```
# Visualization using Seaborn
```

In [212…
```
#encodeing - convert str to numeric
df_visual.columns
```

Out[212…
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'churn_risk', 'tenure_days', 'cancellation_month'],
      dtype='str')
```

In [214…
```
#df_encoded =
df_visual[['plan_type','contract_type','churn_score','churn_flag','churn_risk','esc
```

Out[214…
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>plan_type</th> <th>contract_type</th> <th>churn_score</th> <th>churn_flag</th> <th>churn_risk</th> <th>escalations</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody> </tbody>

| | | | | | |
|---|---|---|---|---|---|
| Standard | Annual | 12 | 0 | low | 0 |
| Premium | Annual | 91 | 1 | high | 1 |
| Basic | Monthly | 34 | 0 | low | 0 |
| Premium | Annual | 8 | 0 | low | 0 |
| Standard | Monthly | 88 | 1 | high | 1 |

In [223…
```
df_encoded = df_visual[['plan_type','contract_type','churn_score','churn_flag','chu

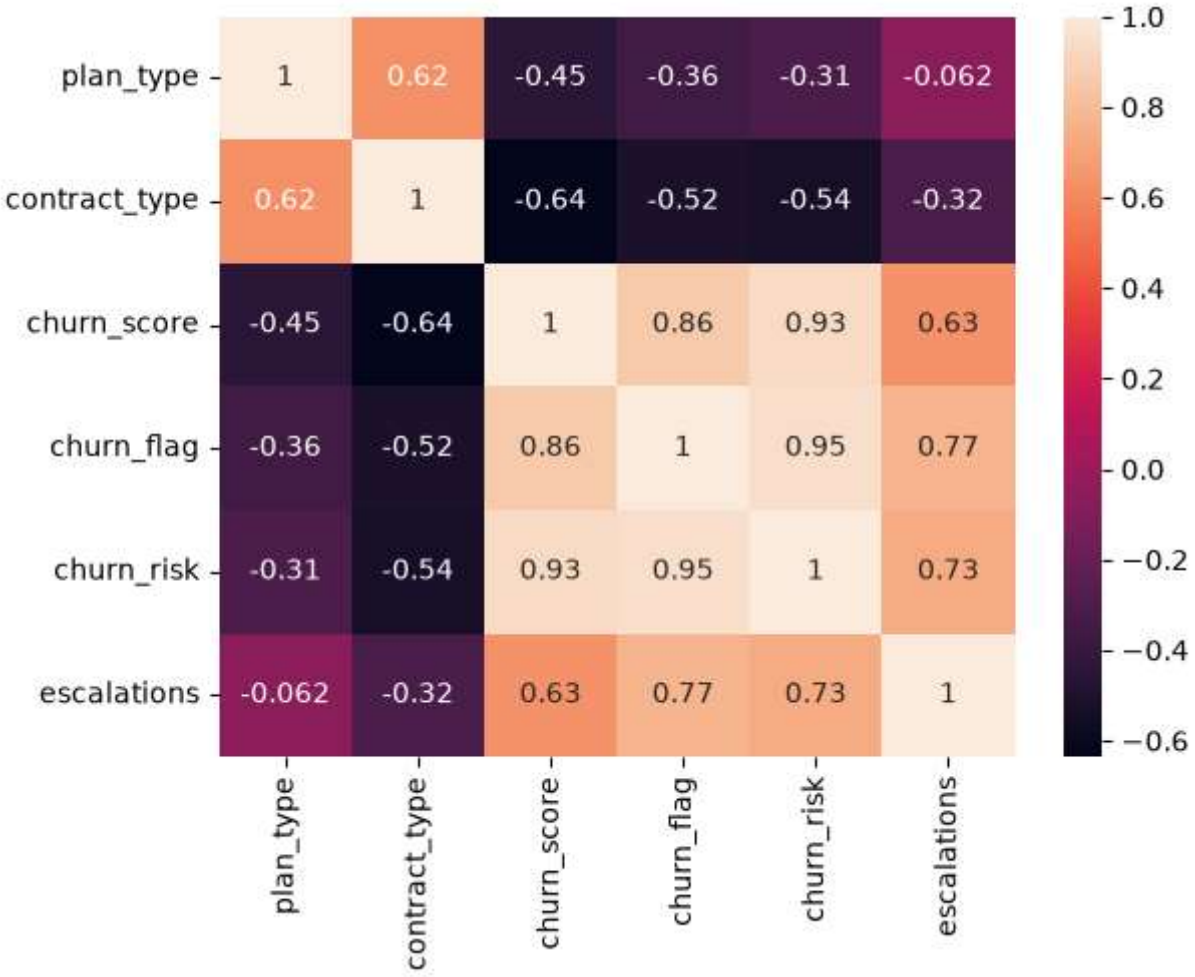
order_mappings = {
    'plan_type':['Basic','Standard','Premium'],
    'contract_type':['Monthly','Annual'],
    'churn_risk':['low','med','high']
     }

for col,order in order_mappings.items():
    df_encoded[col] = pd.Categorical(df_encoded[col].astype('category'), categories
```

In [224…
```
#Heatmap (correlation matrix)

sns.heatmap(df_encoded.corr(), annot=True)
```

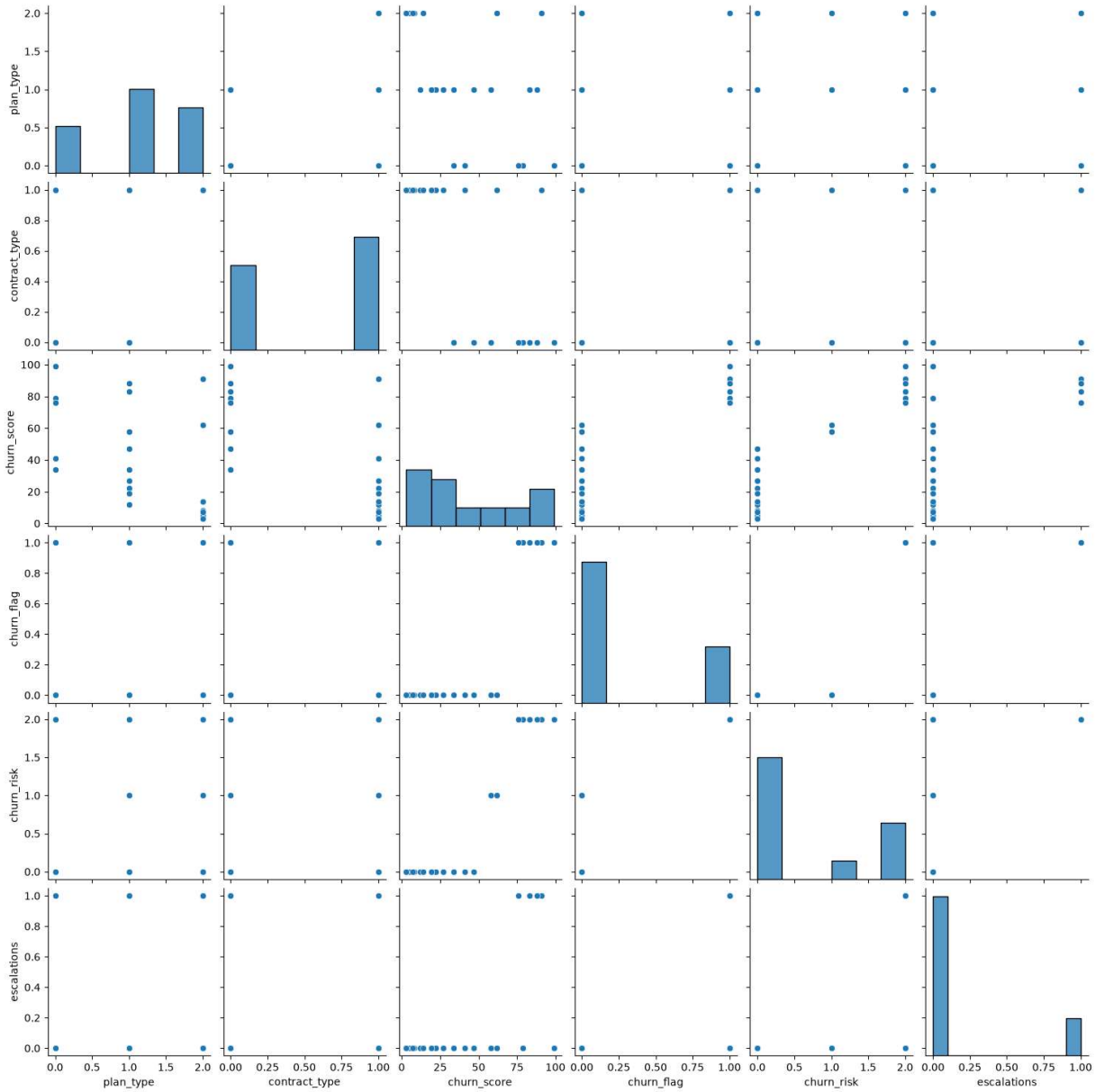
Out[224…
```
<Axes: >
```




In [225…
```
#pairplot

sns.pairplot(df_encoded)
```

Out[225…
```
<seaborn.axisgrid.PairGrid at 0x16a3ad2fe00>
```



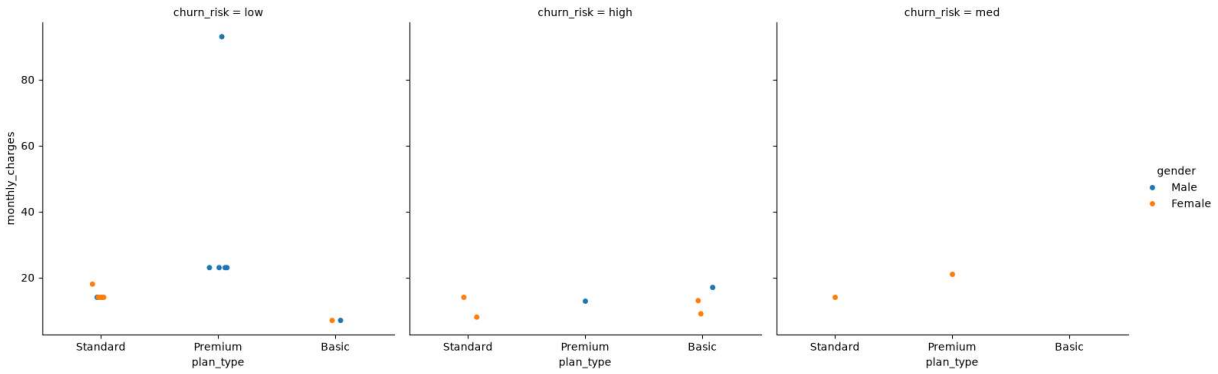

In [227…

```
#catplot/facegrid plot

sns.catplot(data=df_visual,
            x='plan_type',
            y='monthly_charges',
            hue='gender',
            col='churn_risk')
```

Out[227…

```
<seaborn.axisgrid.FacetGrid at 0x16a5a773620>
```




In [228…
```
#pivot table
```

In [230…
```
pd.pivot_table(
    df_visual,
    values='churn_flag',
    index='plan_type',
    aggfunc= 'mean'
).reset_index()
```

Out[230…
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>plan_type</th> <th>churn_flag</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> </tbody> <tbody></tbody>

| | plan_type | churn_flag |
|---|---|---|
| 0 | Basic | 0.600000 |
| 1 | Premium | 0.142857 |
| 2 | Standard | 0.222222 |

In [231…
```
pd.pivot_table(
    df_visual,
    values=['monthly_charges','customerid','churn_flag'],
    index='plan_type',
    aggfunc= {
        'monthly_charges': 'sum',
        'customerid': 'nunique' ,
        'churn_flag': 'mean'
    }
)
```

Out[231…

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>churn_flag</th> <th>customerid</th> <th>monthly_charges</th> <th>plan_type</th> <th></th> <th></th> <th></th> </thead> <tbody> <th>Basic</th> <th>Premium</th> <th>Standard</th> </tbody> <tbody></tbody>

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0.600000 | 5 | 52.95 | | | | | |
| 0.142857 | 7 | 218.93 | | | | | |
| 0.222222 | 9 | 123.91 | | | | | |

In [ ]:

In [232…
```
#working with SQ: in python (pandas)
```

In [233…
```
#create db in sql

conn = sqlite3.connect('test_database.sqlite')

#table details
conn.execute("CREATE TABLE users(first_name TEXT, country TEXT , budget INTEGER)")

#commit and save
conn.commit()
```

In [234…
```
#inser data

cursor = conn.cursor()

#write sql query to insert records in sql table
cursor.execute(
    """
       INSERT INTO users VALUES
          ('Madhav','india',5000),
          ('Rishabh','Germany',2500),
          ('Vishakha','India',3500)
     """
)

#commit and save
conn.commit()

print("Data Inserted Successfylly!")
```

```
Data Inserted Successfylly!
```

In [235…
```
#check inserted data in table
conn = sqlite3.connect('test_database.sqlite')
query = """SELECT * FROM users"""

df_results = pd.read_sql(query,conn)

df_results.head()
```

Out[235…

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>first_name</th> <th>country</th> <th>budget</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> </tbody> <tbody></tbody>

| | | | |
|---|---|---|---|
| | Madhav | india | 5000 |
| | Rishabh | Germany | 2500 |
| | Vishakha | India | 3500 |

In [237…

```
#aggregation
query = """
        SELECT country,sum(budget) as total_budget
        FROM users
        GROUP BY country
        """
df_agg = pd.read_sql(query,conn)
df_agg
```

Out[237…

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th> </th> <th>country</th> <th>total_budget</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> </tbody> <tbody></tbody>

| | | |
|---|---|---|
| Germany | 2500 | |
| India | 3500 | |
| india | 5000 | |

In [238…

```
conn.close()
```

In [ ]: